

8

Market Transparency

Learning Objectives:

- What market transparency is
- Its different dimensions
- How it differs across markets
- How it affects liquidity and price discovery
- The implications for regulation

8.1 Introduction

Securities markets are often taken to be the archetype of transparency, whereby all participants are perfectly informed of the terms of past trades and those at which they could trade at every point in time. In practice, of course, this is rarely the case. Consider a retail investor trying to buy a share of IBM. He could check the latest prices available on the internet, but these are delayed, so that he would still be uncertain about the exact current price. For more accurate information, he could subscribe to a real-time data feed such as Bloomberg or Reuters. But these services are costly, and they indicate the price of the most recent past transaction, not the quotes available for the next trade. To get such quotes, our investor must hire a broker or subscribe to Openbook Ultra, a service that displays all posted limit orders for NYSE stocks.

And the market for IBM is a relatively transparent one. Consider instead an investor who wants to buy or sell a U.S. corporate bond. This is an OTC market, for which data on past trades are available only with a delay of 15 minutes (via a system called TRACE) and no information on dealers' quotes is published: the only way to learn about the price you might get in this market is—again—to contact a broker or your bank's brokerage office, who will in turn inquire about the prices quoted by the various dealers.

This lack of transparency pervades securities markets worldwide. And by no means does it affect only retail investors: even mutual fund managers or brokers often do not have a full picture of the trading process, especially if the relevant market is highly fragmented. For instance, a broker or dealer in an OTC market,

such as the foreign exchange market, cannot possibly know all the opportunities that are available simultaneously. Yet it is important for market professionals to have access to as much information as possible about market conditions, since these professionals quote prices on the market and are crucial to the price discovery process.

The foregoing implies that transparency may refer to different kinds of information, such as the quotes on future trades—pre-trade transparency—or past trades and prices—post-trade transparency. These types of information play different roles. Information on quotes enables market participants to limit execution cost and risk, and fosters competition between liquidity providers, as explained in Section 8.2. Meanwhile, post-trade data help market participants hone their estimates of a security's value and of other participants' strategies, as shown in Section 8.3. Transparency can also extend to the identities of market participants—those involved in past transactions or those currently posting a quote. That is, transparency also encompasses the issue of “anonymity” of trading, an issue raised in Section 8.4.

Transparency is one of the most hotly debated issues in securities market regulation. As explained in Section 8.5, different degrees of transparency mean different distributions of rents across market participants and can even shut some of them out of the market altogether. Moreover, fine-tuning transparency is a crucial choice variable for competing exchanges, as it affects their relative attractiveness to traders. Each market must disclose some data to attract trading, but in doing so it may enable competitors to piggy-back on price discovery.

8.2 Pre-Trade Transparency

Different market structures impose different constraints on pre-trade transparency. The least transparent are OTC markets, such as those for small company stocks, municipal bonds, corporate bonds, or bespoke derivatives like credit default swaps (CDSs). Typically, these are thinly traded securities; no firm quotes are publicly posted, and prices are only available from a dealer upon request. Such markets, where trading interest is at best sporadic, do not repay the time and effort that dealers would have to expend to monitor price-relevant information continuously. At the other extreme, there are the electronic LOB markets for blue chips, where any potential trader can purchase real time information on prices and quantities before placing an order. In between, there is a range of intermediate cases. An example is a dealer market with firm quotes publicly displayed on screen for limited trade sizes, where customers can obtain price improvements by contacting individual dealers.

There are three forms of pre-trade transparency: (i) visibility of quotes, (ii) visibility of incoming orders, and (iii) visibility of traders' identities. These three forms of pre-trade transparency have different effects:

- (i) Visibility of quotes reduces dealers' rents and so enhances liquidity (Section 8.2.1); moreover, it enables customers to fine-tune their orders to the liquidity supply and so reduce execution risk (Section 8.2.2).
- (ii) Visibility of incoming orders helps dealers to detect informed investors, leading to narrower bid-ask spreads and better price discovery (Section 8.2.3).
- (iii) Visibility of order submitters' identities has an ambiguous effect on liquidity: it reduces trading costs for investors who are identified as uninformed, but it may impair liquidity for the others (Section 8.4).

8.2.1 Quote Transparency and Competition between Dealers

This section shows how the lack of pre-trade information on quotes can lessen competition among dealers and thus reduce market liquidity. To see this, consider again the model developed in Chapter 3. There we assumed that dealers' quotes are freely and perfectly observable, so customers costlessly compare quotes and choose the best price. Dealers are accordingly driven to offer zero-profit quotes, and liquidity is maximal. For instance, if there is no asymmetric information, risk aversion, or order-processing cost, the competitive bid and ask quotes are both equal to the expected value of the security, and the spread is zero.

Now suppose instead that dealers' quotes are not visible and clients must contact dealers sequentially to get quotes. On receiving a quote, the investor can either accept it and trade or reject it and contact another dealer. Getting a new quote is costly: investors pay a search cost c for each request of a quote. Each client can be a buyer or a seller with equal probabilities. Buyers are willing to pay up to $\mu + \tau$ (where μ is the expected value of the security) to buy the security, while sellers want to receive at least $\mu - \tau$ for it. Dealers value the security at μ . These differences in valuation might reflect, for instance, different hedging needs (an investor with a long position in a bond, say, is willing to buy a CDS at a markup to its fair value). These differences generate a motivation for trading.

In this setup, the presence of a search cost, however small, enables dealers to charge monopoly prices in equilibrium, that is, bid and ask quotes that are respectively equal to buyers' and sellers' reservation values. To see that this is an equilibrium, suppose that dealers quote these prices. Consider a buyer who is matched with one dealer. The dealer offers to sell at $\mu + \tau$. Since the investor expects other dealers to quote the same price, he has no incentive to

use resources to shop around. Now consider the dealers. The dealer that the investor contacted has no incentive to offer a better price than $\mu + \tau$, since he expects the investor to accept it. Nor do other dealers have any reason to quote a better price: they cannot advertise their quotes, because the market is opaque. Replicating this argument symmetrically on the bid side, the bid price will be $\mu - \tau$, so the bid-ask spread is 2τ instead of zero.

Interestingly, there is no equilibrium except for the monopoly pricing equilibrium. To see this, suppose that this is not so—that is, that there is an equilibrium in which at least one dealer quotes an ask price less than $\mu + \tau$. Denote him as dealer 1 and suppose the investor is matched with this dealer. Dealer 1 can raise his offer by a small amount (up to c) without losing this customer since the client cannot get a better offer (dealer 1 being the cheapest) and would have to pay the search cost c . As dealer 1 has a profitable deviation, this cannot be an equilibrium. This argument has been standard in the literature on industrial organization since Diamond (1971), and extends beyond our specific assumptions regarding the shape of customers' demand and supply curves.

Thus pre-trade opaqueness is conducive to market power for dealers, and hence to less liquidity. Dispersion in dealers' prices could be a symptom of such market power: if investors have different private value estimates (that is, different τ), the above model suggests that the prices they receive will be highly dispersed, as dealers' market power enables them to price discriminate (assuming that they know investors' private valuations). Moreover, different types of investors have different search costs: sophisticated ones such as hedge funds or mutual fund managers have smaller search costs than retail investors thanks to their continuous market presence. One therefore expects sophisticated investors to get better prices, which should result in a less-dispersed distribution of the prices they get. For instance, in the previous model, if there are investors for whom $c = 0$, they all receive a price of μ , regardless of their private valuations.

Evidence from the municipal bond market (SEC 2004; Harris and Piwowar 2006; Green, Hollifield, and Schuerhoff 2007) is consistent with these implications. The trading costs for municipal bonds are substantially higher than for equities, and particularly high for retail-sized rather than institutional trades. This is presumably because retail investors are less sophisticated. Furthermore, the SEC report (2004) shows that prices for retail transactions (below \$10,000) are dispersed: intraday price differences exceeding 3% across dealers are quite common (over 9% of transactions), even though intraday fluctuations in the fundamental value of municipal bonds are minimal. By contrast, there is less price dispersion for large, institutional transactions: for trades of size around \$1 million, a difference exceeding 1% is rare (0.4% of transactions) in the sample. In 2004, the National Association of Securities Dealers (NASD) sanctioned major dealers that had failed to meet their obligation to buy and sell at fair prices,

after finding major discrepancies in pricing: one bond was sold on behalf of a customer for less than half the price that it traded for later in the day; another client received seventy cents on the dollar for a bundle of bonds with a fair value of 97 cents.

8.2.2 Quote Transparency and Execution Risk

Investors do not always observe all the limit orders posted in the market in real time. This is problematic, since there is considerable time variation in the liquidity available in LOBs. This situation creates uncertainty for investors about the prices they may get: they may end up trading at the wrong time (when price impacts are large) or the wrong amount (too much when the LOB is thin). In other words, if the market is opaque, it will be harder for investors to adjust to market conditions, so that on average they will gain less from trading.

A simple example can illustrate these points. Assume that quotes are given by the following schedule, as in Chapter 4:

$$p(q) = \mu + \lambda q, \quad (8.1)$$

where $p(q)$ is the price obtained by a customer who places an order of size q . Unlike the previous chapters, now suppose that the price impact parameter, λ , is random. For instance, it could be high ($\lambda = \lambda^H$) or low ($\lambda = \lambda^L$) with equal probability. We denote the expected value of λ by $E(\lambda)$.

As in the previous section, consider again a buyer who values the security at $\mu + \tau$ and wishes to place a market order. If the market is opaque, when the investor chooses the size q of this order, he does not know the realization of λ , though he correctly anticipates that his order will affect the market price according to price impact function (8.1) and accordingly sets his order size to maximize:

$$\max_q E[(\mu + \tau)q - pq] = \tau q - E(\lambda)q^2.$$

The first order condition of this problem yields the optimal order size, q^O , when the market is opaque:

$$q^O = \frac{\tau}{2E(\lambda)}. \quad (8.2)$$

Thus, the investor optimally trades more when the wedge τ between his valuation of the security and that of liquidity suppliers increases, and less if he expects his price impact to be greater.

In contrast, in a transparent market where the investor knows his price impact, λ , he sets his order size, q^T , to maximize:

$$\max_q \tau q - \lambda q^2.$$

The first order condition to this problem yields the optimal order size:

$$q^T(\lambda) = \frac{\tau}{2\lambda}. \quad (8.3)$$

Hence, when the market is transparent, the investor fine-tunes his strategy to the exact amount of liquidity present in the market, λ . Thus, in contrast to q^O , his order size, $q^T(\lambda)$, depends on the realized value of λ and therefore is itself random. By Jensen's inequality,¹

$$E\left(\frac{1}{\lambda}\right) > \frac{1}{E(\lambda)}. \quad (8.4)$$

Thus, comparing equations (8.2) and (8.3), the average optimal order size is greater in the transparent than in the opaque market:

$$E(q^T(\lambda)) > q^O.$$

The intuitive reason is as follows. The investor trades more in the transparent market when the price impact of his own trade is small (when $\lambda = \lambda^L$) and less when the price impact is large ($\lambda = \lambda^H$). The increase in size in the first case more than offsets the restraint in the second. Hence, on average he ends up trading more.

Interestingly, the investor's expected gain from the trade is also greater in the transparent market. To see this, observe that on average his profit in the transparent market is:

$$E(\tau q^T - \lambda (q^T)^2) = \frac{\tau^2}{4} E\left(\frac{1}{\lambda}\right),$$

whereas in the opaque market, it is:

$$\tau q^O - E(\lambda)(q^O)^2 = \frac{\tau^2}{4} \frac{1}{E(\lambda)}.$$

¹ This inequality states that, for any random variable x , $E[f(x)] > f[E(x)]$ if $f(\cdot)$ is a convex function. Here we apply this principle to the function $1/x$, which is convex.

Again using inequality (8.4), we see that the investor fares better in the transparent market—once again, transparency enables him to make better trading decisions.

These findings carry three implications. First, pre-trade transparency is valuable, so people are willing to pay for real-time quote information or to employ intermediaries who have access to it. Second, pre-trade transparency increases the volume of trade, that is, it encourages participation in the market. Third, if there is some persistence in the illiquidity parameter λ over time, investors have an incentive to make their trades conditional on past measures of market liquidity, since these are informative about current liquidity—a point formalized by Hong and Rady (2002).

8.2.3 Order Flow Transparency

In some markets—foreign exchange or bond markets, to name two—different orders get filled almost simultaneously by different liquidity providers, who may know very little about trades made simultaneously by their competitors. In this situation, the order flow is not transparent. What effects does this have?

To address this issue, we compare two market structures, using a slightly modified version of the framework developed in Chapter 3: (i) a completely opaque dealer market in which each dealer observes only his own order flow and (ii) a transparent market where liquidity providers see all the orders submitted before setting their prices. The latter situation captures the wealth of information available to speculators on the floor of an open-outcry auction market (such as the London Metal Exchange).

As in Chapter 3, we consider the market for a risky security with payoff v that can be high (v^H) or low (v^L) with equal probability, so that the expected payoff is $\mu = (v^H + v^L)/2$. Quotes are set by risk-neutral dealers and are valid for one share. Dealers can be contacted either by informed or uninformed investors. Specifically, with probability π two risk-neutral informed traders are present on the market: they both place a market order to buy one share if they know $v = v^H$ or to sell if $v = v^L$. With probability $1 - \pi$, there are two liquidity traders, one buyer and one seller. These assumptions represent the simplest possible way to capture the idea that informed trading tends to generate positively correlated orders, while uninformed trading does not (the case in which noise trades are not perfectly negatively correlated yields similar conclusions—see exercise 1). Figure 8.1 describes the order arrival process.

Traders randomly contact one of the dealers quoting the best price and can trade with only one dealer at a time. The crucial difference from the model considered in Chapter 3 is that now two trades (not one) may happen simultaneously

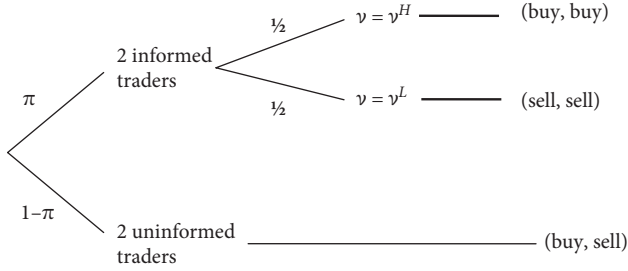


Figure 8.1 Order arrival process in the order flow transparency.

in the market. As noted, in the opaque market dealers only observe the order that comes to them, while in the transparent market they see all orders, whether they receive them or not.

At the time he accepts an order in an opaque market, a dealer does not know the direction of the other order that has come to the market. Hence, his ask price, a^O (where the superscript O stands for “opaque”) is the expected value of the security, conditional only on the fact that he has received a buy order. As π is the probability that the order comes from an informed trader, equation (3.10) in Chapter 3 yields the dealer’s ask price:

$$a^O = \mu + \pi(v^H - \mu). \quad (8.5)$$

The bid price symmetrically is:

$$b^O = \mu - \pi(\mu - v^L), \quad (8.6)$$

so that the bid-ask spread is:

$$s^O = \pi(v^H - v^L). \quad (8.7)$$

In the transparent market, dealers have more information when they set their quotes, because they observe the orders that go to their competitors as well. Hence, their quote is the estimate of the security’s value, *given all the orders submitted*. Under our assumptions, this information reveals whether the traders are informed and tells the direction of their signal. For instance, if two buy orders are submitted, then dealers infer that informed traders know that $v = v^H$ and therefore quote an ask price equal to v^H . The dealers’ valuation and price for each possible configuration is given below, where the superscript T stands for “transparent”:

- (i) two buy orders: $E(v | B, B) = v^H$, so that $a^T = v^H$,
- (ii) one buy and one sell order: $E(v | B, S) = \mu$, so that $a^T = b^T = \mu$,
- (iii) two sell orders: $E(v | S, S) = v^L$, so that $b^T = v^L$,

The average price paid or received by liquidity traders is μ , which means that the bid-ask spread for them is zero. In this example, transparency eliminates their trading cost. Hence, again in this case, opaqueness increases the trading costs for liquidity traders. The reason is that with transparency dealers detect informed traders more easily, so they can charge a lower spread to the uninformed. This inverse relationship between transparency and uninformed investors' trading costs holds more generally, as Pagano and Röell (1996) show. In the transparent market, by contrast, informed investors obtain no profit since they must always pay the maximum bid-ask spread, $v^H - v^L$. As a result, the average spread in the transparent market is $\pi(v^H - v^L)$, as in the opaque market. In the transparent market, the allocation of trading costs differs between uninformed and informed investors as transparency—unlike opacity—allows dealers to discriminate between the two groups.

In this model, the lower trading costs for liquidity traders mirror a decrease in trading profits for the informed trader, since trading is a zero-sum game and dealers make zero expected profits. Thus a change in market transparency can redistribute gains between different types of participants. This explains why transparency is such a controversial issue.

Order flow transparency also has implications for price discovery. To see this, let us use the average squared pricing error as an inverse measure of price discovery. Let p^O denote the price paid to or by a dealer in the opaque market. This will be a^O if a dealer executes a buy market order and b^O if he executes a sell market order. In the opaque market, the average squared pricing error is:

$$\begin{aligned} E[(p^O - v)^2] &= \frac{\pi}{2}(a^O - v^H)^2 + \frac{\pi}{2}(b^O - v^L)^2 \\ &\quad + \frac{1-\pi}{2} \left[\frac{1}{2}(a^O - v^H)^2 + \frac{1}{2}(b^O - v^H)^2 \right] \\ &\quad + \frac{1-\pi}{2} \left[\frac{1}{2}(a^O - v^L)^2 + \frac{1}{2}(b^O - v^L)^2 \right]. \end{aligned}$$

The first two terms correspond to the cases in which the dealer trades with an informed investor (receiving good news in the first case and bad news in the second). The third term corresponds to the case in which the dealer receives an order from an uninformed investor and the true value of the security is high. The last term corresponds to the symmetric situation of an order from an uninformed investor when the true value is low. Substituting out for the ask price a^O from (8.5) and the bid price b^O from (8.6), this expression simplifies to:

$$E[(p^O - v)^2] = (1 - \pi^2)(v^H - \mu)^2,$$

which is decreasing in π : as the frequency of informed trading increases, the price discovery process improves.

In the case of the transparent market, the transaction price p^T is either equal to the true value if there are informed investors (with probability π), or equal to μ if there are uninformed traders (with probability $1 - \pi$). Hence, the average squared pricing error is:

$$E[(p^T - v)^2] = (1 - \pi) \left[\frac{1}{2}(v^H - \mu)^2 + \frac{1}{2}(v^L - \mu)^2 \right] = (1 - \pi)(v^H - \mu)^2.$$

This pricing error is clearly smaller than the corresponding expression in the opaque market, the difference between the squared pricing errors being:

$$E[(p^O - v)^2] - E[(p^T - v)^2] = \pi(1 - \pi)(v^H - \mu)^2 > 0.$$

Thus, transparency also improves price discovery because it helps market participants to learn about the presence of informed traders.

8.3 Post-Trade Transparency

Another issue regarding transparency is the timeliness of disclosure of past trades. In some markets, information is released in real time to all market participants (for a fee); in others, past trades are disclosed with a significant delay, if ever. The speed of publication of information on past trades is one of the most controversial issues in the organization of securities markets. Dealers often oppose prompt disclosure on the ground that it makes it more difficult for them to manage their inventory (and therefore raises inventory holding costs). In this section, we use a variant of the previous section's model to show how the lack of post-trade transparency may enable dealers to capture rents at the expense of other traders. That is, we have another instance of transparency redistributing profits and costs.

Assume that the orders are filled in sequence over two periods, $t = 1$ and $t = 2$, as in Figure 8.2. With probability π , two successive identical orders are placed by informed investors and executed at their arrival times, $t = 1$ and $t = 2$. With probability $1 - \pi$, two orders of opposite signs are placed by uninformed traders in random sequence.

Consider first a market with post-trade transparency: dealers have to report their trades as they take place, and the exchange publishes the information

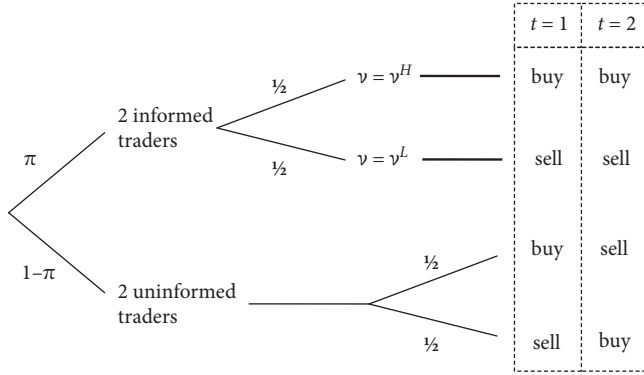


Figure 8.2 Order arrival process in the post-trade transparency model.

immediately. In this case, the quotes posted by dealers at time $t = 1$ are the same as those prevailing in the opaque market analyzed in the previous section, since in the first period dealers have no information on forthcoming orders. Hence, in the first period, the bid-ask spread is:

$$s_1^T = \pi(v^H - v^L). \quad (8.8)$$

In contrast, at time $t = 2$, the dealer who receives the second order is in the same position as in the transparent market analyzed before: he and all his peers observe his new order, and they have all learned about the previous trade. Consider first the case in which the first-period order was a buy. If another buy order arrives at time $t = 2$, dealers infer that the investor is informed and $v = v^H$. Hence, their ask price is equal to v^H . If instead, a sell order arrives, they infer that the investor is uninformed, so $v = \mu$ and their bid price is μ . Hence, following a buy order in the first period, the bid-ask spread in the second period is $v^H - \mu = (v^H - v^L)/2$. If the first-period order is a sell, the analysis is symmetric. Thus, the bid-ask spread posted in the second period is:

$$s_2^T = \frac{1}{2}(v^H - v^L).$$

To summarize, with post-trade transparency, the spread will vary over time, and trading costs for uninformed and informed investors will differ. For instance, liquidity traders pay a spread of s_1^T in the first period, and zero in the second. Summed over the two periods, the average trading cost of uninformed investors in the post-trade transparent market is:

$$TC^T = \frac{1}{2} \times s_1^T + \frac{1}{2} \times 0 = \frac{\pi}{2}(v^H - v^L). \quad (8.9)$$

This shows that post-trade transparency is an imperfect substitute for pre-trade transparency: on average, over the two periods, the trading cost for uninformed traders is intermediate between what they would pay with pre-trade transparency and pre-trade opacity.

Let us now compare the outcome with and without post-trade transparency. Where it is absent, first-period transactions are not disclosed before second-period trading. In this case, in the second period the dealer who has already received an order in the first period has an informational advantage: if it was a buy order, he knows that the value of the security is high with a greater probability than other dealers; with a sell order, he knows that it is low. In general, he will exploit this advantage by adjusting his quotes to capture profitable trading opportunities that his peers are not aware of.

To see this as simply as possible, assume that in the second period dealers set their quotes sequentially and that the dealer who received the first-period order goes last.² This dealer can always undercut any competitor's quote that could yield a positive expected profit and refrain from undercutting those that are loss-making. This is an extreme instance of the well-known "winner's curse" problem in auction theory: if you win the trade, it can only be because better informed bidders know that the price you are paying is already too high! In response to this problem, the best quotes that competing dealers can offer without losing money are the highest and lowest possible values that the informed dealer may place on the security, i.e., $a = v^H$ and $b = v^L$.

Now consider the best response of the dealer who executed the order arriving at date $t = 1$. Assume first that it was a sell order. If another sell order comes in, the dealer knows that the security has a low value so that his bid price is v^L , like that of the other dealers. In this case, if he executes, he makes no profit. But if a buy order comes in next, he knows that there is no informed trading and so he estimates the security to be worth μ . He will then slightly undercut the ask price of v^H set by his competitors. Thus, he will get to execute the buy order (as he offers the best price) and will make a profit almost equal to $v^H - \mu$. Therefore, the only scenario in which a dealer makes money in the second period is a sequence of two orders in opposite directions. This only happens with probability $1 - \pi$, since it occurs if and only if the two orders are placed by liquidity traders. That

² This dealer behaves as a "Stackelberg follower." Though unrealistic, this assumption ensures the existence of a pure-strategy equilibrium in the price-setting game between asymmetrically informed dealers. If instead dealers were assumed to set their quotes simultaneously, the equilibrium is necessarily a mixed-strategy one. This has been shown for first-price common value auctions with differentially informed bidders, to which our game is equivalent (see Engelbrecht-Wiggans, Milgrom, and Weber 1983). For a treatment of the problem in our setting, see Röell (1988). See exercise 3 as well.

is, the dealer who manages to capture the first-period trade earns an expected informational rent in the second period of $(1-\pi)(v^H - \mu)$.

This expected second-period gain comes at the expense of investors, since other dealers are careful to set their quotes so as to make zero profits. Hence, the second-period trading costs are as high as they can possibly be in our setup: the second-period spread is always:

$$s_2^O = v^H - v^L.$$

This result shows that lack of post-trade transparency, which increases informational asymmetries among dealers, is a source of informational rents and thereby impairs market liquidity.

At the same time, however, the prospect of such informational rents can sharpen first-period competition between dealers to capture order flow and the attendant information.³ In our setting, first-period quotes will be driven down to levels such that the total expected profit of each dealer over both periods is nil. The implied first-period ask price is obtained by subtracting the second-period informational rent $(1-\pi)(v^H - \mu)$ from expression (8.5):

$$a_1^O = \mu + \pi(v^H - \mu) - (1-\pi)(v^H - \mu) = \mu + (2\pi - 1)(v^H - \mu). \quad (8.10)$$

The bid price is symmetric, so that the first-period spread is:

$$s_1^O = (2\pi - 1)(v^H - v^L), \quad (8.11)$$

which is clearly smaller than the second-period spread computed above: $s_1^O < s_2^O$. Therefore, in a market with post-trade opacity, the time profile of the bid-ask spread is rising: initially dealers accept low spreads and incur losses, which they subsequently recoup by higher spreads. This pattern for the bid-ask spread has been found experimentally by Bloomfield and O'Hara (1999, 2000).⁴ Interestingly, dealers often argue that post-trade opacity enables them to offer better quotes to their clients. This is in fact the case in the first period of the model, but the practice is really just a way to prepare the ground for much less competitive quotes in subsequent periods.

Note that if π is low enough (less than $\frac{1}{2}$), the model predicts that with post-trade opacity, the first-period spread can be negative! This may seem

³ In this respect, dealers have been compared to the bookmaker who declared that he was happy to lose money to a particularly successful bettor: "He is my most valuable client. I always shorten the odds when he bets and he saves me a fortune" (*Financial Times*, December 6, 1987).

⁴ Most experimental work regarding pre-trade transparency has found that markets with greater pre-trade transparency feature lower bid-ask spreads. See for instance Flood et al. 1999.

paradoxical, although such “crossed quotes” (or “locked quotes”) do sometimes arise briefly in some markets. However, it is a situation that cannot last, since it is a clear opportunity for arbitrage profit—you can buy a security and resell it at the same time for a higher price! If arbitrage prevents a negative bid-ask spread, then dealers will retain strictly positive expected profits over the two periods, which could explain why they generally oppose post-trade transparency. And their hostility to transparency will be further reinforced if first-period competition is too weak to fully dissipate their overall profit.

Who gains and who loses from post-trade opacity? If price competition among dealers in the first period drives their overall expected profits to zero, all we need to do to find out is compute the total trading costs of the liquidity traders in the opaque market. We get:

$$TC^O = \frac{1}{2}s_1^O + \frac{1}{2}s_2^O = \pi(v^H - v^L), \quad (8.12)$$

which is double the trading cost under post-trade transparency (see equation (8.9)). Thus, post-trade opacity ultimately raises informed traders’ expected profits at the expense of liquidity traders.⁵ The reason is that in the opaque market the problem of adverse selection persists into the second period, whereas in the transparent market it is eliminated by last-trade publication at the end of the first period. More precisely, in the opaque market adverse selection persists because a dealer gleans from the first-period order flow information that enables him to gain a competitive edge, so as to both outbid other dealers and exploit this edge at the expense of liquidity traders.

This model highlights the fact that dealers may wish to acquire order flow information, either with smaller spreads or through direct monetary inducements such as payment for order flow (i.e., compensation offered to brokers by a dealer in return for channeling orders to him). In the model, these rebates dissipate the informational rents created by opaqueness. In reality, dealers may be able to set them so as to keep some rents. Thus, both dealers and informed traders could benefit from post-trade opacity at the expense of noise traders. This again illustrates how changes in transparency redistribute trading profits and costs among market participants.

As for price discovery, it can be shown that it is less efficient in the opaque market (see exercise 2). Intuitively, in the opaque market, quotes underadjust relative to the information contained in the order flow. In the first period, dealers

⁵ At first glance, this is surprising since competition among dealers in the first period dissipates their rents. Hence one would expect uninformed investors to entirely recoup their second period losses with lower trading costs in the first period. But this is not the case, because informed investors capture part of the benefits of the lower spread in the first period.

are willing to execute a buy order at a price below their current value estimate, conditional on receiving a buy order. But in the second period, dealers set quotes that are independent of the first-period order and that are therefore totally uninformative, generating a discrepancy between transaction prices and fair value. In contrast, in the transparent market, second-period quotes are perfectly informative.

8.4 Revealing Trading Motives

Transparency need not be limited to prices and quantities but may extend to the identity of potential counterparties. This information can be price relevant insofar as it offers insight into the reasons why people want to trade, and specifically whether they have superior information about the value of the security. For example, consider a broker who receives a large sell order from a mutual fund facing substantial customer redemptions. He may get a better price for his client if he can reassure market makers that, given its source, the order is most unlikely to be driven by superior information.

Market designs differ in the scope for conveying such detailed information about trading motives. For example, floor markets allow participants to see one another and interpret subtle cues like impatience and nervousness, as well as to voluntarily share information about their clients. Similarly, dealer markets enable brokers to interact with dealers by phone and explain their clients' reason for trading. This exchange of pre-trade information can be seen as a way of increasing pre-trade transparency for the parties involved.⁶ Whether it improves the overall transparency of the market depends on how many people are included in the exchange of information. Knowing their motivations enables liquidity suppliers to offer different prices to different traders, opening up the possibility of price discrimination.

Transmission of information about trading motives—and the attendant price discrimination—is less easy in electronic LOB exchanges, where liquidity providers must post quotes without observing the identities of the brokers who are placing market orders. Of course, there are still ways of getting around the problem. If the exchange reveals the codes of the limit order submitters, which not all exchanges do, then a broker could reveal his identity by placing a marketable limit order at the best existing quotes. But the broker's identity provides at best imperfect guidance about the motives for the order.

⁶ In principle, the exchange of information need not be only from traders to liquidity suppliers, as is assumed in this section. For instance, market makers may also reveal their identities to traders who place orders with them, or they may not—an issue touched on in the next section.

Alternatively, the broker may call up counterparties to arrange a transaction before clearing it in the main market—but this takes time and effort.

In some cases, the need to disclose the trading motives and the identities of the liquidity demanders is so great that non-anonymous trading mechanisms emerge, not necessarily at the exchange authorities' initiative. For instance, an investor could publicly announce a trade and its planned execution date some days in advance, a practice is called "sunshine trading." To this purpose, in 1997, the Swiss stock exchange (SIX Swiss Exchange) created a "second trading line," where firms can buy their own shares back openly on a separate market segment, to avoid exacerbating adverse selection in the main market—firms being generally considered well informed about their own prospects (Chung, Išakov, and Pérignon 2007).

How does the exchange of information about trading motives affect liquidity? To answer, we resort once more to the one-period model presented in Chapter 3, where π and $1-\pi$ are the probabilities of an order coming from an informed and a liquidity trader, respectively. But we now assume that before trading, a fraction κ_U of the uninformed traders is recognized as such, perhaps because they manage to reveal their lack of information via a sunshine trading announcement.⁷ Symmetrically, a fraction κ_I of the informed traders is also found out, possibly because they have developed a reputation of trading in advance of news. An important assumption, for the moment, is that a trader's identity, if revealed, is observed by all market makers. This assumption will later be relaxed. The order arrival process is summarized in Figure 8.3.

Now, a trader who places a buy order will face a different ask price, depending on whether he is recognized or not; if he is recognized, he will pay a different

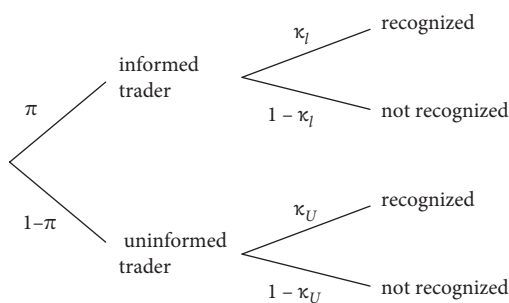


Figure 8.3 Order arrival and recognition of trading motives.

⁷ For models that study how investors can signal their trading motives, see Seppi (1990) or Admati and Pfleiderer (1991) for instance.

price depending on his type. The easiest cases are those in which traders' motives are recognized. If the buyer is known to be informed, the market makers' best value estimate is v_H , while if he is known to be uninformed, it remains at its unconditional value μ . Now consider the case in which the trader's type is unknown. We need to compute the probability that this trader is informed, conditional on not having been identified. This probability is:

$$\pi' = \frac{\pi(1 - \kappa_I)}{\pi(1 - \kappa_I) + (1 - \pi)(1 - \kappa_U)},$$

that is, the fraction of informed traders within the population of unidentified traders. As this probability plays the same role as π in the model of Chapter 3, we deduce that the ask and bid prices posted for an order of unknown origin are:

$$\begin{aligned} a &= \mu + \frac{\pi(1 - \kappa_I)}{\pi(1 - \kappa_I) + (1 - \pi)(1 - \kappa_U)}(v^H - \mu), \\ b &= \mu - \frac{\pi(1 - \kappa_I)}{\pi(1 - \kappa_I) + (1 - \pi)(1 - \kappa_U)}(v^H - \mu). \end{aligned} \quad (8.13)$$

As expected, the revelation of information on trading motives results in price discrimination. Recognized uninformed traders trade at a zero spread, since they can buy or sell the security at price μ , while informed traders identified as such trade at the fair price and therefore see their trading profits eliminated. Last, all unrecognized traders, regardless of type, trade at the pooled, partially informative prices given in equations (8.13).

The model is open to different interpretations for different values of κ_I and κ_U . The case of $\kappa_U > 0$ and $\kappa_I = 0$ can be interpreted as the coexistence of two populations of uninformed traders: some are identified as such and obtain a zero spread, because they can access the upstairs market or practice sunshine trading; others are pooled with informed traders. Clearly, uninformed traders in the former group benefit from being recognized, while the others are disadvantaged because they face a larger bid-ask spread than they would under anonymous trading. The bid-ask spread faced by unidentified traders is:

$$s = \frac{\pi}{\pi + (1 - \pi)(1 - \kappa_U)}(v^H - v^L),$$

which increases in κ_U and is therefore minimal for $\kappa_U = 0$. The reason is that the pool of unidentified traders now has a smaller proportion of uninformed traders than in an anonymous market. However, the average uninformed trader is better off, since only a proportion $1 - \kappa_U$ pay this spread s , and the average spread $(1 - \kappa_U)s$ decreases with κ_U . Intuitively, since there are fewer uninformed traders

in the main market, the expected profits of the informed traders will decrease. So reducing anonymity increases the posted bid-ask spread, even though it reduces average trading costs for uninformed traders.

Consider now the opposite case in which some informed traders—but no uninformed traders—can be recognized, that is $\kappa_U = 0$ but $\kappa_I > 0$. This can be thought of as a situation in which regulation obliges some potentially informed traders to disclose their intentions. For example, a potential takeover raider may be forced to disclose that the fraction of the target company's equity (his "toehold") has breached some regulatory threshold (often 5% of the company's voting equity): if subsequent large buy orders are attributed by market participants to such an investor, they should have a larger price impact.

In this case, the bid-ask spread for unidentified traders is:

$$s' = \frac{\pi(1 - \kappa_I)}{\pi(1 - \kappa_I) + (1 - \pi)}(v^H - v^L),$$

which is decreasing in κ_I : identifying informed traders improves market liquidity.

This analysis suggests that disclosing trading motives benefits uninformed traders, at least on average, but the conclusion rests on a crucial assumption: namely, that information on trading motives is disclosed to *all* market participants. If it were revealed only to a subset of price setters, the conclusion might be different. To see this, in the context of the model with $\kappa_U > 0$ and $\kappa_I = 0$, suppose that the customer's type is observed only by the dealer he contacts, possibly owing to a long-standing relationship that confers credibility on the information revealed by the customer—a situation that cannot be replicated with other dealers.

This situation gives the dealer market power, as he can appropriate some or all of the informational rent involved in the transaction. To see this, suppose that a buyer is identified by his dealer as uninformed. The best price that this buyer can then expect from the dealer is μ , which corresponds to zero spread and zero profit for the dealer. At the opposite extreme, the worst possible price that he will accept from the dealer is his outside option $\mu + s/2$, the ask price that he would get in the main market. In this situation of bilateral monopoly, the relative bargaining power of the buyer and the dealer determines which price within the interval $[\mu, \mu + s/2]$ the traders actually settle upon. If the client has some bargaining power, he negotiates a discount from $\mu + s/2$, his main market price. Such price improvements were in fact quite common in dealer markets, such as Nasdaq and the LSE (Reiss and Werner 2004).

In the extreme case in which the dealer has all the bargaining power, the client gets the same price as unrecognized traders who buy on the main market, that is $\mu + s/2$. As a result, all traders will be strictly worse off than under

completely anonymous trading, since s is smallest when $\kappa_U = 0$. In fact, it is easy to verify that this remains true whenever the dealer's bargaining power is great enough that the discount is low. This shows that revealing trading motives may not lower the trading costs of uninformed traders (even those whose motives are revealed) if the information remains confidential (disclosed to just one dealer).

Channeling the orders of uninformed clients to specific dealers or markets is known as cream-skimming, the practice for which brokers often obtain a payment for order flow. If brokers also have a market-making capability (i.e., they are broker-dealers), they may skim off the uninformed orders for in-house execution, a practice known as internalization, modelled by Röell (1990) and Battalio and Holden (2001).

There is evidence that cream skimming does take place: Easley, Kiefer, O'Hara, and Paperman (1996b) find that orders diverted from the NYSE are less informative than those that remain on the NYSE. Grammig and Theissen (2012) obtain the same result for orders internalized on Deutsche Börse's Xetra trading system. They also find that investors whose orders are internalized do not capture the entire benefit of lower adverse selection costs. Since skimming deprives the main market of a proportion κ_U of the uninformed order flow, it should reduce market liquidity according to this model. However, the evidence is mixed (Battalio 1997; and Battalio, Greene, and Jennings 1998).

We deal with internalization, payment for order flow, and cream-skimming when discussing market fragmentation in Chapter 7. Questions about these practices resurface here, because opacity and fragmentation are intertwined. For instance, order flow transparency (discussed in Section 8.2.3) is harder to achieve when traders can split market orders among multiple trading platforms, unless there are extremely efficient linkages among them.

8.5 Why Are Markets So Opaque?

The previous sections lead to the conclusion that transparency about quotes, orders, and traders' identities generally enhances market liquidity, at least as far as uninformed traders are concerned. The natural question, then, is why we see so little of it: why are so many real-world securities markets so opaque along one or more of the dimensions we have analyzed. A simple explanation may be that these market structures are not designed to benefit uninformed traders but market-making intermediaries or informed traders (see Section 8.5.1). Indeed there is good reason to think that these market professionals have far greater influence in the design of trading rules than uninformed and occasional traders.

Even if market making is competitive, so that it yields no rents to intermediaries, the incentive to offer opacity to large trades still exists, even though this

may decrease the liquidity available to retail traders. This issue is examined in Section 8.5.2.

But the opacity of securities markets may also reflect concerns about economic efficiency, at least in some cases. For instance, transparency may be problematic—it may even reduce liquidity—if it exposes limit-order placers to a high risk of being picked off by more informed traders. Moreover, if markets are opaque, market makers may find it more difficult to sustain collusive agreements because it will be harder to detect and punish violations of the cartel's rules. In this sense, opacity may foster competition between market makers. Section 8.5.3 discusses these brighter sides of market opacity.

8.5.1 Rent Extraction and Lobbying

As Section 8.2.1 shows, market makers can extract rents in a market with little pre-trade transparency on quotes, and informed traders can do so in one with little post-trade transparency. In some cases these two categories actually coincide: intermediaries may have superior information (partly due to their advance knowledge of the order flow), and so they may resist making their quotes visible, in part to retain informational rents.⁸

This is consistent with the empirical evidence that transaction costs fall when markets become more transparent. For example, the bid-ask spread on the NYSE declined when the exchange started releasing information about the LOB of its listed stocks (Boehmer, Saar, and Yu 2005); similarly, investors' trading costs in the corporate bond market dropped substantially after the TRACE trade reporting system was implemented in 2002 (Bessembinder and Maxwell 2008). And even in such inherently opaque markets as real estate or travel services, the introduction of web-based search engines has reduced the profits of incumbent firms.

Trading platforms themselves can also extract rents from opacity by restricting access to data on prices and trades, thus increasing the value of the data they sell, even to the point of provoking complaints from some participants. For instance, the NYSE's decision to charge a fee for real time information on quotes and trades in Archipelago (a trading platform acquired in 2006) stirred up strong opposition from investors. More generally, the pricing of market data has become very controversial, as investors need information about quotes

⁸ In Bolton, Santos, and Scheinkman (2016), opaque asset markets ("OTC markets") enable informed intermediaries to sustain abnormal profits. Moreover, informed intermediaries cream-skim high quality assets, worsening the quality of the pool of assets that trade on exchanges (transparent markets). This generates a negative externality, which leads to excessive intermediation in equilibrium.

and trades to make optimal routing decisions in fragmented markets.⁹ Buy side investors complain that trading platforms charge too high prices for their data, while exchanges respond that they incur costs that they need to cover to disseminate those data. Revenues from sales of market data account for an increasing share of exchange revenues. For instance, revenues from data sales for Nasdaq increased from \$413 million in 2014 to \$779 million in 2019 (including co-location revenues).

A way in which trading platforms can squeeze even more profits from the data that they generate is by selling “low-latency” (super-fast) access to data feeds, even granting to some traders the opportunity to co-locate their computers in physical proximity to the platform’s own computer. Such preferential high-speed access is keenly sought by algorithmic traders, who can make money by beating their competitors even by nanoseconds. This again creates tiered access to market data, and enables high-frequency traders to reap large informational rents at the expense of other market participants, as explained in Chapter 9.

Besides enabling intermediaries to retain rents from market power and informed traders to extract profits from superior information, opacity may allow brokers to cheat their customers by misreporting trade prices, or to simply conceal the fact that they exerted little effort or care to make sure their customer got the best price available on the market at the time of execution. This may occur even where best-execution rules are in place, because enforcing such rules is harder in opaque markets: investors are less likely to discern that they are poorly served by their brokers, and to assemble the evidence necessary to demonstrate dishonesty or negligence.

Therefore, in general market professionals should be expected to favor limiting the transparency of the markets where they operate. And it is no surprise that their interests influence market regulation more than those of other market users. Not only do the professionals have a more regular market presence, but nowadays they largely own, control, and manage stock exchanges and multilateral trading facilities, as a result of the trend towards demutualization (described in Chapter 2).

The resistance of market professionals (the “sell side” of the market) to raising transparency standards in security trading can explain why regulatory reforms with this aim often have disappointing outcomes. This was the case, for instance, for the MiFID 2 directive in 2018, which was explicitly aimed at raising the transparency of security markets in the EU (see Section 7.5.2). A survey of data prices in Europe, carried out by the European Securities and Markets

⁹ This is an enduring issue: Mulherin, Netter, and Overdhal (1991) provide a fascinating historical account of how the NYSE asserted its property rights over prices set on the NYSE in the nineteenth century.

Authority (ESMA) in 2019, found that the 2018 implementation of MiFID 2 had not resulted in market data being supplied to the market on a “reasonable commercial basis,” as the regulation had stipulated. Moreover, a consultation conducted in 2020 by the ESMA found that all market participants shared the view that the reform failed to achieve its objective to create meaningful transparency for non-equity markets. However, interestingly, respondents had split views on whether transparency standards should be raised further to overcome this failure: a handful of sell-side firms and trade associations objected to it, emphasizing the role played by banks and other intermediaries in putting their capital at risk, while a much larger group of respondents, including trading venues and proprietary traders, broadly supported it—consistent with the idea that raising transparency standards tends to hurt liquidity suppliers and benefit liquidity demanders.

8.5.2 Opacity Can Withstand Competition

The previous section may suggest that there is an intrinsic connection between opacity and rent extraction (or collusion) by trading intermediaries. If so, one might expect that competition from new entrants, unaffiliated with incumbents, would spark a race towards transparency. But in fact opaque markets are resilient to competition.

To see why, consider again the model of post-trade transparency presented in Section 8.3. Suppose that dealers are allowed to choose whether to publicize their trades. In this case the only equilibrium is such that none chooses to publicize trades. The reason is that the market maker who does not publicize trades can undercut those who do, because he can use the information gleaned from the first-period trades to make a profit in the second period. To see this, recall that the competitive bid-ask spread at time $t = 1$ in the opaque market (equation (8.11)) is lower than the spread that a transparent market maker could charge (equation (8.8)). Thus, in that model all first-period trading will be captured by opaque market makers. Note that the choice about the degree of opacity can be made not only by individual dealers but also at the level of trading platforms on which they operate. Under this interpretation, what the model predicts is that competing platforms will inevitably opt for opacity in the design of their trading system, and such platforms will capture all trading.

In reality, opaque market makers or platforms will compete aggressively only for large orders: small trades, which are unlikely to be informative, will be of little or no interest, since they can be filled just as well by transparent market makers.

This leads to a two-tier market, where large trades execute opaquely and small ones transparently.

The foregoing implies that market forces alone are unlikely to produce post-trade transparency, at least for large trades. Regulation is required to mandate a minimum level of transparency. For instance, the U.S. corporate bond market was opaque until 2002, when the SEC mandated the TRACE system to disseminate price and trade information, albeit with a lag.

Even mandatory disclosure, however, encounters considerable problems of enforcement. First, there may be regulatory arbitrage: if trading can migrate to venues that do not impose less post-trade transparency, it will. Many observers of the migration of the wholesale markets in European blue-chip equities to SEAQ International in London in the early 1990s argued that a primary determinant was the lack of trade publication there. Europe-wide negotiations on market regulation and transparency typically pitted the looser, more free-wheeling regime of the London wholesale markets against the more centralized, transparent regime prevailing in the home markets. Nowadays, the increasing role of dark pools and of the OTC market in the United States and post-MiFID Europe can be explained using the same logic.

Second, it may be possible to evade post-trade transparency requirements. Franks and Schaefer (1995) point out that trade reporting deadlines can be circumvented by using “protected trades”: market makers informally offer a price to a customer for a large deal (with the understanding that they will, in practice, honor the commitment), but the deal is not officially finalized and confirmed until they have had time to unwind the resulting inventory. This delays the trade report—and its publication—beyond the regulators’ intention. Moreover, it reduces immediacy and imposes some execution risk on large traders.

8.5.3 The Bright Side of Opacity

Is the self-interest of market professionals the only explanation for the prevalence of opaque trading systems? The fact that even the most transparent marketplaces allow for some opaque forms of order placement suggests otherwise. For instance, most electronic order book systems—otherwise known for their high degree of pre-trade transparency—allow traders to enter hidden limit orders or iceberg orders. In this case, only a small portion of the order is visible to other traders, and the rest gradually becomes visible as the order executes against incoming orders. Of course, other investors will eventually

realize when a hidden order is present on the LOB, and adapt their trading strategies accordingly (De Winne and D'Hondt 2004). This takes time, however, and even then the total size of the order remains unknown.

One reason why an investor may want to place such an order is to avoid giving away his private information to the market by placing a visible large order.¹⁰ This motivation would be in line with the idea that opacity benefits informed traders. A trader may also consider placing a hidden order to protect himself against the risk of being picked off by better informed traders, if he is afraid, say, that his information may be (or may become) stale and his order accordingly mispriced. Indeed, placing a buy limit order is tantamount to offering a free put option to other market participants (Copeland and Galai 1983); likewise, a sell limit order is a free call option. Investors who receive new information can exploit such an option by hitting existing limit orders before they can be cancelled (see Chapter 9). Therefore, traders who submit limit orders bear an adverse selection cost similar to that facing market makers: the difference is that with limit orders, the possible losses to informed trading are more limited, as limit orders vanish once hit (and they are typically hit as soon as they go into the money), while dealer quotes remain available for trading until the dealer manages to update them.¹¹

Therefore, if uninformed investors could not place hidden orders, they would most likely place market orders, possibly trickling them into the market to minimize price impact. As a result, the LOB would be less deep than with hidden orders, which in practice do provide a considerable amount of liquidity. For example, in an active LOB such as that for the CAC 40 blue-chip French stocks on average, the displayed depth at the best five quotes accounts for less than 55% of the total depth at these prices, that is, more than 45% is provided by hidden orders (De Winne and D'Hondt 2004). Here, then, we have an instance in which imposing a greater degree of transparency (banning hidden orders) would be likely to reduce liquidity, by discouraging the use of limit orders.

A similar argument holds for anonymity, again insofar as it applies to limit-order traders, who provide liquidity, rather than to market-order traders, who consume it. Recall again the analogy between limit orders and options: as the value of an option depends on the volatility of the underlying security, the value of a limit order to the market—and hence the cost to its placer—increases with volatility. Hence in anticipation of increased volatility, traders should be less willing to place limit orders, or at least bid less aggressively to reduce the

¹⁰ A cost however is that hidden orders have a lower execution probability and longer time to completion: see Bessembinder, Panayides, and Venkataraman (2009).

¹¹ Equivalently, they can be seen as auction participants exposed to the winner's curse problem, because their buy limit orders are more likely to be filled when they overestimate the true value, and their sell orders are more likely to be filled when they underestimate it (see Chapter 6).

risk of being picked off. As a result, the bid-ask spread should widen. So a wide bid-ask spread signals that limit-order traders expect high volatility. Now suppose that some traders know future volatility and others do not: the latter can infer volatility from the limit order book, which also contains offers posted by the informed traders. But if trading is anonymous, uninformed traders can no longer pick out the limit orders from informed participants. So, when they see a wide spread, they don't know precisely whether or not it reflects high future volatility.

In this setting, Foucault, Moinas, and Theissen (2007) show that if the fraction of informed traders is high, the uninformed will tend to see a large spread as a hint that future volatility is high. They will cave in, thus contributing to a large bid-ask spread. But if their fraction is low, the uninformed traders will consider a large spread as relatively uninformative, and so will trade aggressively, thereby tending to narrow the spread. So in the first scenario anonymity increases the spread, while in the second it reduces it. They show empirically that the latter result is consistent with the data from Euronext (the French Stock Exchange), where identifiers for the brokers placing limit orders ceased to be disclosed after April 23, 2001. After this date, the average quoted spread became significantly smaller; in this case, anonymity increased liquidity. Notice that this result does not contradict those of Section 8.4, which refer to the anonymity of market-order placers, not to that of limit-order placers.

The anonymity of quotes may also deter collusion between quote setters, or at least make it more complicated. When identities of quote setters are unknown, it may be difficult or impossible to “punish” quote setters who violate previous accords, so quote setting is likely to become more competitive. This is another channel through which pre-trade opacity may increase liquidity: Simaan, Weaver, and Whitcomb (2003) find that market makers are more likely to quote on odd ticks and actively narrow the spread when they can do so anonymously by posting limit orders on ECNs. Thus, from a public policy perspective, decreasing the level of pre-trade transparency by allowing anonymous quotes could improve price competition and narrow spreads.

8.6 Further Reading

In this chapter, we have mainly analyzed transparency in models of trading with asymmetric information. However, the literature on pre-trade quote transparency also covers situations where market-maker pricing is determined by inventory holding costs. Biais (1993) posits that market makers differ in inventories (and thus in their inventory holding costs) but are otherwise approximately risk neutral and in agreement on the fundamental value. He compares

two market structures: a transparent market in which dealer inventories (or equivalently quotes) are observed by all market participants and an opaque market in which dealers have no information on the inventories (quotes) of their competitors. Effectively, the former is an English auction and the latter is a sealed-bid auction, and the two produce different market maker bidding strategies. The revenue equivalence theorem of auction theory applies to this setting (risk neutral bidders with independent, identically distributed private values) so that, as Biais concludes, the expected spread will be the same in the two markets.

Beyond this stylized setting, auction theory states that revenue equivalence breaks down if the bidders (the market makers) are risk averse, if their private valuations are correlated or asymmetrically distributed, or if there is an imperfectly known common value generating a winner's curse problem (see, e.g., Bolton and Dewatripont 2004). Within Biais's framework, de Frutos and Manzano (2002) relax the assumption of low market-maker risk aversion and conclude that the opaque market is more liquid. Yin (2005) extends the analysis by Biais by incorporating the idea that quote transparency exerts competitive pressure by eliminating customers' search costs. He considers the case in which customers must pay a search cost in the opaque market and finds that the average spread is smaller in the transparent market.

A few empirical studies have examined how a change in pre-trade transparency affects market quality. Hendershott and Jones (2005) consider the impact of a SEC regulatory action that prompted the Island ECN to stop displaying limit orders in three of the most actively traded ETFs for about a year. They find that this was associated with a drop in liquidity for these ETFs. Boehmer, Saar, and Yu (2005) find that liquidity increased when the NYSE started releasing information on the LOBs of its listed stocks in 2002, as predicted by Baruch (2005). However, Madhavan, Porter, and Weaver (2005) consider a similar change on the Toronto Stock Exchange and obtain the opposite conclusion.

The empirical evidence on post trade transparency is also mixed. Gemmill (1996) considers three different reporting regimes (no publication, 90 minutes delay, and 24 hours delay) for dealers on the LSE, and finds no significant impact on liquidity. In contrast, more recent empirical work (Bessembinder, Maxwell, and Venkataraman 2006; Edwards, Harris, and Piwowar 2007; and Goldstein, Hotchkiss, and Sirri 2007) suggests that the implementation of post-trade transparency in the U.S. corporate bond market led to a significant fall in trading costs. Röell (1988, 1995) models post-trade transparency and finds that post-trade transparency benefits retail uninformed traders in particular.

Foucault, Pagano, and Röell (2010) offer a brief survey of the theoretical and empirical literature on transparency.

8.7 Exercises

1. **Uncorrelated noise traders' orders.** Modify the setting of Section 8.2.3 by assuming that if noise traders are present, each independently chooses to place either a buy or a sell order. In other words, we can have the following orders by noise traders: (buy, buy), (buy, sell), (sell, buy), (sell, sell), each occurring with probability $(1-\pi)/4$. Compute the bid and ask prices under pre-trade transparency and opaqueness. How do the predictions of the model change?
2. **Price discovery and transparency.** Consider the model of post-trade transparency described in Section 8.3. Consider the time-averaged expected squared deviation between the transaction price and the true value of the security, that is:

$$\frac{E[(p_1^k - v)^2]}{2} + \frac{E[(p_2^k - v)^2]}{2},$$

where p_t^k is the transaction price in period $t = 1, 2$ in regime $k = T, O$ (transparent, opaque). Show that price discovery is more efficient in the transparent market. You may limit your analysis to the case $\pi > \frac{1}{2}$ in which the equilibrium first-period spread is positive.

3. **Asymmetric information and dealers.** (Hard question!) Analyze the opaque regime of the model described in Section 8.3 under the assumption that in the second period the market maker who has inherited order flow information from the first period posts his quotes simultaneously with the other market makers—that is, that there is no Stackelberg leader in the second period. First, provide an intuitive argument that there is no pure-strategy equilibrium of the quote-setting game among market makers in the second period. Then, characterize the ensuing mixed-strategy equilibrium.